Yeeun Kim, Ph.D.

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EDUCATION

Korea Advanced Institute of Science and Technology Daejeon, South Korea
Ph.D., Department of Civil and Environmental Engineering Aug. 2023

- Dissertation: *Dynamic multilevel traffic simulation for urban traffic considering human driver behavior* (Advisor: Dr. Hwasoo Yeo)

Korea Advanced Institute of Science and Technology Daejeon, South Korea
M.S., Department of Civil and Environmental Engineering Feb. 2018

Korea Advanced Institute of Science and Technology Daejeon, South Korea
B.S., Department of Civil and Environmental Engineering Feb. 2016

ACADEMIC APPOINTMENTS

University of Central Florida Orlando, FL, U.S.A.
Postdoctoral Researcher, Smart and Safe Transportation Lab. (Advisor: Dr. Mohamed Abdel-Aty)
Mar. 2025 – Present

Korea Advanced Institute of Science and Technology Daejeon, South Korea
Postdoctoral Researcher, AIxMobility Lab. (Advisor: Dr. Hwasoo Yeo) Sep. 2023 – Feb. 2025

École Polytechnique Fédérale de Lausanne Lausanne, Switzerland
Visiting Scholar, Urban Transport Systems Lab. (Advisor: Dr. Nikolas Geroliminis) Sep. 2022 – Dec. 2022

- Funded by NRF and ETH Zurich (Young Researchers' Exchange Program between South Korea and Switzerland 2022)

New York University Abu Dhabi Abu Dhabi, UAE
Visiting scholar Jan. 2019 – Dec. 2019

PUBLICATIONS

Peer-Reviewed Journal Articles

** corresponding author. Flagship-venue metrics shown as (JCR quartile, latest JCR IF, Scimago 2026 h-index).*

[J1] **Yeeun Kim***, Mohamed Abdel-Aty, and Uibeom Chun (2026). Exploring Factors Influencing Speeding Patterns With Connected Vehicle Data for Road Context-Sensitive Policy Implications. *Transportation Research Part A*, 211, 105098. https://doi.org/10.1016/j.tra.2026.105098 **(Q1, IF 7.2, h-index 199)**

[J2] **Yeeun Kim**, Seongjin Choi, Sujae Jeon, and Hwasoo Yeo* (2026). NextSim: Multi-Level Traffic Simulation for Urban Networks Using Dynamic Level Assignment. *IEEE Transactions on Intelligent Transportation Systems*, 27(7), 8564-8578. https://doi.org/10.1109/TITS.2026.3671220 **(Q1, IF 9.1, h-index 244)**

[J3] Hyunsik Min, Gyeongseon Baek, **Yeeun Kim**, and Byeongjoon Noh* (2026). Anticipatory Driving Maneuver for Safety and Comfort: Temporal Graph Cross Attention-Based Front Vehicle Trajectory Prediction. *IEEE Transactions on Intelligent Transportation Systems*, Early Access. https://doi.org/10.1109/TITS.2026.3703652 **(Q1, IF 9.1, h-index 244)**

[J4] Jihu Kim, **Yeeun Kim**, and Hwasoo Yeo* (2026). Section-Based Crash Risk Analysis Integrating the Effect of Traffic States and Road Geometry. *IET Intelligent Transport Systems*, 20(1), e70134. https://doi.org/10.1049/itr2.70134 **(Q2, IF 2.7, h-index 74)**

[J5] Sujae Jeon, **Yeeun Kim**, and Hwasoo Yeo* (2025). A Two-Dimensional Lane-Changing Model With Asymmetric Driving Behavior. *Journal of Korean Society of Transportation*, 43(6), 815-829. https://doi.org/10.7470/jkst.2025.43.6.815

[J6] Hyunsik Min, **Yeeun Kim**, Jungryul Kim, Seongjin Lee, and Byeongjoon Noh* (2025). Anticipating Front-Side Vehicles' Maneuvers for Trustworthy Autonomous Driving. *The Journal of the Korea Institute of Intelligent Transport Systems*, 24(6), 121-133. https://doi.org/10.12815/kits.2025.24.6.121

[J7] **Yeeun Kim** and Hwasoo Yeo* (2024). Asymmetric Repulsive Force Model: A New Car-Following Model With Psycho-Physical Characteristics. *Transportation Research Part C: Emerging Technologies*, 161, 104571. https://doi.org/10.1016/j.trc.2024.104571 **(Q1, IF 8.4, h-index 214)**

[J8] **Yeeun Kim**, Hye-young Tak, Sunghoon Kim, and Hwasoo Yeo* (2024). A Hybrid Approach of Traffic Simulation and Machine Learning Techniques for Enhancing Real-Time Traffic Prediction. *Transportation Research Part C: Emerging Technologies*, 160, 104490. https://doi.org/10.1016/j.trc.2024.104490 **(Q1, IF 8.4, h-index 214)**

[J9] **Yeeun Kim**, Sunghoon Kim, and Hwasoo Yeo* (2021). A Study on Traffic Prediction Using Hybrid Approach of Machine Learning and Simulation Techniques. *The Journal of the Korea Institute of Intelligent Transport Systems*, 20(5), 100-112. https://doi.org/10.12815/kits.2021.20.5.100

[J10] Minju Park, **Yeeun Kim**, and Hwasoo Yeo* (2019). Development of an Asymmetric Car-Following Model and Simulation Validation. *IEEE Transactions on Intelligent Transportation Systems*, 21(8), 3513-3524. https://doi.org/10.1109/TITS.2019.2930320 **(Q1, IF 9.1, h-index 244)**

[J11] **Yeeun Kim**, Seongjin Choi, and Hwasoo Yeo* (2019). A Study on Development of a Car-Following Model for Accident Simulation Caused by Driver Distraction. *Journal of Korean Society of Transportation*, 37(1), 39-50. https://doi.org/10.7470/jkst.2019.37.1.039

[J12] **Yeeun Kim**, Young-Ji Byon, and Hwasoo Yeo* (2018). Enhancing Healthcare Accessibility Measurements Using GIS: A Case Study in Seoul, Korea. *PLoS ONE*, 13(2), e0193013. https://doi.org/10.1371/journal.pone.0193013 **(Q1, IF 2.8, h-index 500)**

[J13] **Yeeun Kim**, Sehyun Tak, Jeongyun Kim, and Hwasoo Yeo* (2017). Evaluation of Accident Prevention Performance of Vision and Radar Sensor for Major Accident Scenarios in Intersection. *The Journal of the Korea Institute of Intelligent Transport Systems*, 16(5), 96-108. https://doi.org/10.12815/kits.2017.16.5.96

Manuscripts Under Review

[R1] Uibeom Chun, **Yeeun Kim***, and Mohamed Abdel-Aty. A Driving Simulator Experiment of Drivers' Response to Roadside Visual Elements Across Suburban and Urban Roadway Environments *(Transportation Research Part F: Traffic Psychology and Behaviour, submitted)*

[R2] **Yeeun Kim***, and Mohamed Abdel-Aty, Heterogeneous Road Environment Effects on Speeding across Driving Behavior Types Identified from Connected Vehicle Data *(Accident Analysis & Prevention, submitted)*

[R3] Uibeom Chun*, **Yeeun Kim**, and Mohamed Abdel-Aty, Driver Distraction Tendency and Safety-Critical Response Heterogeneity: Implications for Personalized Forward Collision Warnings *(IEEE Open Journal of Intelligent Transportation Systems, submitted)*

[R4] **Yeeun Kim**, Mohamed Abdel-Aty, and Hwasoo Yeo*, Modeling Human-Like Car-Following Behavior: A Literature Review and Future Research Directions *(Transport Reviews, submitted)*

[R5] Joobin Jin, Seokjun Hong, Gyeongseon Baek, **Yeeun Kim**, and Byeongjoon Noh*, Ctx2TrajGen: Traffic Context-Aware Microscale Vehicle Trajectories Using Generative Adversarial Imitation Learning *(IEEE Transactions on Intelligent Transportation Systems*, *under review)*

[R6] Gyeongseon Baek, Dongho Ka, Jinseong Kim, **Yeeun Kim**, and Byeongjoon Noh*, Lane-Changing Imminence-Aware Trajectory Prediction via Spatio-Temporal Interaction Context Modeling for Mixed Traffic *(IEEE Transactions on Intelligent Transportation Systems*, *under review)*

SELECTED PRESENTATIONS

** corresponding author.*

[C1] **Yeeun Kim*** and Mohamed Abdel-Aty. Driver-Type Heterogeneity in Road Environment Effects on Speeding Behavior. 106th Transportation Research Board Annual Meeting, 2027 (accepted for presentation).

[C2] Uibeom Chun, **Yeeun Kim***, and Mohamed Abdel-Aty. A Driving Simulator Experiment of Drivers' Response to Roadside Visual Elements Across Suburban and Urban Roadway Environments. 106th Transportation Research Board Annual Meeting, 2027 (accepted for presentation).

[C3] Sujae Jeon, Yeonwoo Yu, Dongheon Lee, Hwasoo Yeo, and **Yeeun Kim***. Bus-UCTM: A Bus-Extended Multi-Lane Urban Cell Transmission Model for Mesoscopic Transit Operational Analysis. 106th Transportation Research Board Annual Meeting, 2027 (accepted for presentation).

[C4] Hyejin Kim*, Mohamed Abdel-Aty, and **Yeeun Kim**. Associating Signalized Intersections Skewness and Connected Vehicle-Based Risky Driving Behavior in Crash Type-Specific Frequency Models. 106th Transportation Research Board Annual Meeting, 2027 (accepted for presentation).

[C5] Uibeom Chun*, **Yeeun Kim**, and Mohamed Abdel-Aty. How Driver Distraction Tendency Shapes the Perception–Response Process and Its Implication for Personalized Forward Collision Warning. 106th Transportation Research Board Annual Meeting, 2027 (accepted for presentation).

[C6] Sujae Jeon, **Yeeun Kim**, and Hwasoo Yeo*. A Two-Dimensional Lane-Changing Model With Asymmetric Driving Behavior for Autonomous Vehicles. 28th IEEE International Conference on Intelligent Transportation Systems (ITSC), 2025.

[C7] **Yeeun Kim**, Seongjin Choi, and Hwasoo Yeo*. Extended Urban Cell Transmission Model Using Agent-Based Modeling. 11th International Conference on Ambient Systems, Networks and Technologies (ANT), 2020.

[C8] **Yeeun Kim**, Sehyun Tak, Jeongyun Kim, and Hwasoo Yeo*. Identifying Major Accident Scenarios in Intersection and Evaluation of Collision Warning System. 20th IEEE International Conference on Intelligent Transportation Systems (ITSC), 2017.

RESEARCH EXPERIENCE

University of Central Florida Orlando, FL, U.S.A.

Evaluation of Drivers' Speed, Attention, and Vigilance (Florida DOT, 2025-2026).

- Led the project and conducted the main experiments, including the driving simulator experiment and connected vehicle data analysis of driver speeding behavior. Results published in Transportation Research Part A.

Korea Advanced Institute of Science and Technology Daejeon, South Korea

Development of Next-Generation AI-Integrated Mobility Simulation and Prediction/Application Technologies for Metropolitan Cities (Ministry of Science and ICT (MSIT), South Korea, 2023-2024).

- Co-led the project and led the proposal writing.
- Designed the modeling framework for NextSim, a micro-meso hybrid traffic simulation, and developed the driver behavior modeling for urban networks. Results published in IEEE Transactions on Intelligent Transportation Systems, filed as a Korean patent application, and validated on a real-world network in Bucheon.

Acquisition of Large-Scale Microscopic Traffic Data in Urban Areas Using a Swarm of Drones (KAIST College of Engineering Global Initiative, 2022).

- Wrote the proposal, managed the project, and designed the drone-based data-collection experiment.

Development of Behavior Model for Cooperative Autonomous Vehicle Simulation and Demand-Responsive Routing Algorithm (Ministry of Land, Infrastructure and Transport (MOLIT), South Korea, 2020-2021).

- Designed and developed the traffic simulation model for testing the DRT algorithm. Results published in Transportation Research Part C.

A Development of Future Traffic Prediction Algorithm by Hybridizing Machine Learning and Simulation Methods (National Research Foundation of Korea (NRF), South Korea, 2020).

- Designed and developed a traffic prediction method combining traffic simulation and machine learning. Results published in Transportation Research Part C.

Development of Cloud-Based Traffic Prediction Simulation Software for Solving Urban Traffic Problems (Ministry of Science and ICT (MSIT), South Korea, 2018-2019).

- Designed and developed an agent-based mesoscopic traffic simulation, applied to a four-district network in Gangnam, Seoul.

Development of Shared Autonomous Vehicle-Based Traffic System for Smart City (Korea Agency for Infrastructure Technology Advancement (KAIA), South Korea, 2017).

- Analyzed travel patterns in the target area.

Multi-Level Safety System for Intersection Collision Prevention (Korea Evaluation Institute of Industrial Technology (KEIT), South Korea, 2016-2017).

- Developed intersection traffic-accident scenarios and analyzed camera- and radar-based collision-avoidance systems. Findings presented at IEEE ITSC.

GRANTS & FELLOWSHIPS

Young Researchers' Exchange Program between South Korea and Switzerland 2022

Funded by NRF and ETH Zurich ($ 7,000)

TEACHING EXPERIENCE

Korea Advanced Institute of Science and Technology Daejeon, South Korea

Lead In-Class Instructor, Department of Civil and Environmental Engineering Spring 2024

- Introduction to Transportation Systems Engineering (CE350, Undergraduate Course)

Guest Lecturer

- (Spring 2024, Spring 2023) Transportation System Analysis and Operations (CE547, Graduate Course)

Teaching Assistant

- (Fall 2021) Special Topics in Smart Convergence<Smart mobility for sustainable society> (CoE491(B), Undergraduate Course)
- (Spring 2020, Spring 2017) Introduction to Transportation Systems Engineering (CE350, Undergraduate Course)
- (Spring 2020) Transportation System Analysis and Operations (CE547, Graduate Course)
- (Fall 2017) Introduction to Geomatics Engineering (CE291, Undergraduate Course)

New York University Abu Dhabi Abu Dhabi, UAE

Teaching Assistant

- (Fall 2019) Selected Topics in Information and Computational Systems – Machine Learning (ENGR-UH 4560, Undergraduate Course)
- (Spring 2019) Advanced Topics in Traffic Modeling, Control and Simulation with AI (ENGR-GH-7910, Graduate Course)

NSF International Research Experiences for Students

US-KOREA Advanced Transportation Infrastructure Informatics Institute (ATI3) Daejeon, South Korea

Teaching Assistant 2022 – 2023

- Artificial Intelligence in Smart Transportation

Guest Lecturer 2024

- Traffic Prediction Using Traffic Simulation and AI

National Research Foundation of Korea International Research Experiences for Students

Korea-US International Cooperation Education Program in the Field of Mobility and Artificial Intelligence: Mobility-Related Research Incorporating Artificial Intelligence Technology Daejeon, South Korea

Teaching Assistant, Program Management 2024 – 2025

- Mobility-related research incorporating artificial intelligence technology

INVITED LECTURES

Exploring factors influencing speeding patterns identified from connected vehicle data considering context classification Aug. 14, 2025

KAIST, Daejeon, South Korea

Development and Use of Traffic Simulation for Advanced Mobility Nov. 21, 2024

Korea University, Sejong, South Korea

HONORS & AWARDS

Best Paper Award 2026
Korean Society of Transportation
Best Paper Award-3rd 2022
US-KOREA Advanced Transportation Infrastructure Informatics Institute (ATI3) Program
Best Paper Award 2021
2021 Spring Conference of the Korea Institute of Intelligent Transport Systems
Seoul Creative Encouragement Awards 2017
Seoul Research Papers Competition Knowledge Management Division 2017

PATENTS

Patents in South Korea

10-2025-0053321 (2025.04.23, Application, South Korea)

- Microscopic-Mesoscopic Hybrid Traffic Simulation with Dynamic Simulation Level Assignment for Large-Scale Urban Networks

10-2843362, (Issued August 1, 2025, Co-inventor: Hwasoo Yeo, South Korea)

- Method for Traffic Simulation Considering Driver's Behavior and Road Section Risk and Safety-vulnerable Region Prediction System

10-2022-0141594 (2022.10.28, Application, South Korea)

- System and Method for Predicting Traffic in Real-time

10-2018-0141857 (2018.11.16, Application, South Korea)

- Traffic Simulator for Verification of ITS System

Software Registration in South Korea

C2023-055435

- Microscopic Traffic Simulation Program for Assessing Impact of Public Transportation for Vulnerable

C2018-039177

- Mesoscopic Traffic Simulation Program for Estimating Traffic States in Urban Areas

ACADEMIC SERVICE

Reviewer

Journals

IEEE Transactions on Intelligent Transportation Systems; Transportation Research Interdisciplinary Perspectives; Transport Policy; Journal of Traffic and Transportation Engineering (English Edition); Traffic Safety Research (KoROAD); Computers; Sensors; Applied Sciences; Sustainability; Indian Journal of Science and Technology

Conferences

IEEE International Conference on Intelligent Transportation Systems; IEEE Intelligent Vehicles Symposium; Transportation Research Board Annual Meeting